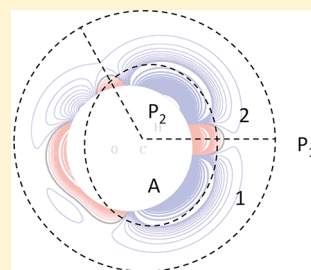


Statistical Theory for the Kinetics and Dynamics of Roaming Reactions

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ABSTRACT: We present a statistical theory for the effect of roaming pathways on product branching fractions in both unimolecular and bimolecular reactions. The analysis employs a separation into three distinct steps: (i) the formation of weakly interacting fragments in the long-range/van der Waals region of the potential via either partial decomposition (for unimolecular reactants) or partial association (for bimolecular reactants), (ii) the roaming step, which involves the reorientation of the fragments from one region of the long-range potential to another, and (iii) the abstraction, addition, and/or decomposition from the long-range region to yield final products. The branching between the roaming induced channel(s) and other channels is obtained from a steady-state kinetic analysis for the two (or more) intermediates in the long-range region of the potential. This statistical theory for the roaming-induced product branching is illustrated through explicit comparisons with reduced dimension trajectory simulations for the decompositions of H_2CO , CH_3CHO , CH_3OOH , and CH_3CCH . These calculations employ high-accuracy analytic potentials obtained from fits to wide-ranging CASPT2 ab initio electronic structure calculations. The transition-state fluxes for the statistical theory calculations are obtained from generalizations of the variable reaction coordinate transition state theory approach. In each instance, at low energy the statistical analysis accurately reproduces the branching obtained from the trajectory simulations. At higher energies, e.g., above 1 kcal/mol, increasingly large discrepancies arise, apparently due to a dynamical biasing toward continued decomposition of the incipient molecular fragments (for unimolecular reactions). Overall, the statistical theory based kinetic analysis is found to provide a useful framework for interpreting the factors that determine the significance of roaming pathways in varying chemical environments.



1. INTRODUCTION

Studies of the photodissociation of formaldehyde and acetaldehyde have demonstrated the existence of a novel mechanism for formation of molecular products.^{1–3} This roaming radical mechanism involves the partial decomposition of a molecule into two weakly interacting radicals, followed by, first, a large-amplitude reorientation of the radicals at separations of about 3 Å (6 au) and, then, a barrierless radical–radical abstraction to produce two closed shell molecular products.⁴ For the formaldehyde and acetaldehyde decompositions these closed shell molecular products are also accessible via standard three-center tight transition states, with the saddle points for these tight transition states lying below the energy of the separated radicals. These two pathways, which are generally well-separated both geometrically and dynamically,^{5,6} produce molecular products with distinct distributions of rovibrational states. The branching between the two pathways is a sensitive function of the energy of the tight transition state relative to the radical asymptote as well as of the dynamics of the roaming process.

Prior descriptions of the “roaming mechanism” have emphasized the presence of multiple pathways to the same products and the resulting production of distinct rovibrational distributions, as outlined in the preceding text.⁷ We prefer an alternative definition that emphasizes the physical nature of the long-range reorientational aspect of the dynamics.⁴ In particular, we define a roaming reaction as one where orientational dynamics (roaming) in the long-range region of the potential (e.g., 2–4 Å separation where dispersion, electrostatic, hydrogen bonding, and/or ion–molecule forces are comparable to chemical forces)

leads to a set of products that is different from those expected by simple continuation of the incipient molecular decomposition (or alternatively of the incipient bond formation or abstraction if considering a bimolecular reaction instead of a unimolecular decomposition). In essence, we consider a roaming reaction to simply be one that produces an alternative set of products via reorientational motion in the long-range region of the potential.

With this definition, the presence or absence of an alternative tight transition-state pathway to closed shell molecular products is irrelevant. Recently, we have demonstrated the ubiquitous nature of the roaming radical mechanism via detailed ab initio studies of the roaming transition-state and minimum energy paths for propane, *n*-butane, isobutane, and neopentane.⁸ Interestingly, for these alkane decompositions any tight saddle point pathways, if they exist, are very high in energy and thus play essentially no role in the kinetics. Nevertheless, roaming pathways clearly exist.

Importantly, our definition in terms of the key aspect of the dynamics provides a unifying concept for a variety of related reactions. In particular, we then have roaming radical,^{1,2,4,9,10} roaming ion–molecule,^{11,12} and roaming radical–molecule^{13–15} reactions as subclasses of the general roaming reaction mechanism. Furthermore, the long-range intermediates that are involved in the roaming may arise from collisions of bimolecular reactants or from partial dissociation of unimolecular reactants. Notably, the

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kinetic scheme that we develop here is readily applicable to each of these reactions.

The effect of the roaming mechanism on the branching between different products is an important kinetic question. For example, the branching between radical and molecular products in a thermal dissociation can have a significant impact on combustion properties such as the ignition delay.¹⁶ In recent work, we have developed a reduced dimensional trajectory approach for predicting this branching. Its initial application to the dissociation of acetaldehyde⁴ yielded predictions for the branching ratios that were in good agreement with the values obtained in shock-tube based experimental measurements.¹⁷ We have subsequently applied this approach to a joint theory/experiment study of the dissociation of dimethyl ether.¹⁸ In this case, while both theory and experiment do suggest some contribution from a roaming pathway, there is a discrepancy in the predicted and observed magnitude that is not well understood.

An even more important aspect of our general definition for roaming is that it leads us to a statistical theory based framework for describing the roaming kinetics and dynamics. Statistical theories have a long history of providing a valuable framework for understanding the factors that affect the kinetics of chemical reactions. Originally, they were largely employed as a means for extrapolating and interpolating experimental data. More recently, the high accuracy of current *a priori* implementations of statistical theories, as shown through numerous theory–experiment comparisons, has led to their direct use as a tool for improving global kinetic models.^{19,20} Unfortunately, it has been presumed that statistical theories are not applicable to the roaming mechanism.^{3,7,21–24}

In this paper, we develop a statistical theory based framework for describing the kinetics and dynamics of reactions that include roaming processes and illustrate its range of applicability with sample studies of the dissociations of H_2CO , CH_3CHO , and CH_3OOH . We also consider a different type of roaming reaction with the competition between dissociation and isomerization in C_3H_4 . The basis of this statistical approach is an assumed separation of the decomposition process into multiple steps. The first step involves a partial decomposition to two weakly interacting fragments. We then presume a statistical competition between the dissociation, isomerization, and return to reactants of these weakly interacting fragments. A steady-state kinetic analysis for the long-range intermediates yields simple expressions for the rate constants and branching ratios in terms of the transition-state fluxes for each of these processes. Prior observations of the presence of two distinct transition-state regions in a wide variety of recombination reactions (including ion–molecule, radical–molecule, and radical–radical reactions)^{25–28} provides strong motivation for this kinetic scheme. Prior comparisons of transition state theory (TST) predictions with experiment also indicate the appropriateness of the reduced dimensional scheme employed here in the treatment of the long-range dynamics.^{4,29}

For concreteness, the statistical framework is first presented and applied here from the perspective of the kinetics and dynamics of dissociation, as given in preceding text. However, it is equally applicable to the bimolecular kinetics of two radicals, and general derivations for that case are also presented. Overall, the present statistical theory framework provides a useful perspective on the kinetics of the roaming process. The consideration of various limiting cases illustrates the factors affecting the relative importance of roaming for different molecules.

The applications indicate that, for low energies, the statistical predictions are in good agreement with the dynamical results.

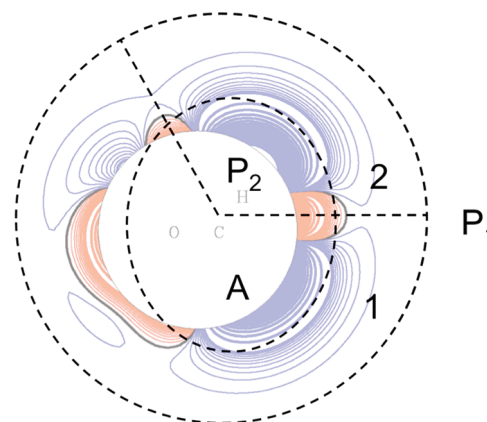


Figure 1. Schematic plot of the dividing surfaces (black dashed lines) separating reactants A (H_2CO), the two long-range intermediates 1 and 2, and the two products P_1 ($\text{HCO} + \text{H}$) and P_2 ($\text{H}_2 + \text{CO}$) from the dissociation of formaldehyde. These dividing surfaces are overlaid on the potential contours for the interaction of an H atom with a rigid HCO radical. The blue contours denote attractive interactions, while the red denote repulsive interactions. The contour spacings are 0.1 kcal/mol for the thin lines and 1.0 kcal/mol for the thick lines.

At higher energies, above about 1 kcal/mol, the statistical predictions increasingly overestimate the dynamical results, due apparently to a dynamical bias toward continued dissociation of the molecules that reach the weakly interacting region. However, such high energies are of reduced importance since the high-energy roaming branching is small anyways. Interestingly, a number of studies of anion–molecule bimolecular kinetics by Hase and co-workers have indicated a closely related direct component whose contribution also increases with increasing energy.^{30–33}

Some earlier ion–molecule studies of Hase and co-workers bring out another important point related to the present model.^{34,35} In particular, a central aspect of the present scheme involves the assumed separation of modes into the intramolecular (or conserved) and intermolecular (or transitional) modes of the fragments. Hase's studies have demonstrated that the coupling between these two sets of modes is very weak in the long-range minimum where the reorientational motions occur. In essence, these studies validate the present approach, which employs a direct treatment of the transitional mode dynamics in the long-range region coupled with vibrationally adiabatic assumptions for the conserved modes. Importantly, the present dynamical tests of the statistical model only consider the dynamics within the transitional mode subspace.

The basic kinetic scheme is derived in section 2. The procedures we employ for the evaluation of the transition-state partition functions and for the trajectory simulations are described in section 3. The potential energy surfaces employed here are summarized in section 4. Then, the results of our illustrative applications are presented and discussed in section 5, with some concluding remarks provided in section 6.

2. STATISTICAL THEORY FOR ROAMING

2.1. Kinetic Scheme for Dissociation of H_2CO . For concreteness we begin by developing our kinetic picture specifically for the dissociation of formaldehyde. A contour plot of the interaction of the H atom with a rigid HCO radical is provided in

Figure 1. Superimposed on this plot are sample dividing surfaces that separate the configuration space into five distinct species. The region labeled A corresponds to the H_2CO species, that labeled P_1 corresponds to separated radical products $\text{H} + \text{HCO}$, and that labeled P_2 corresponds to separated molecular products $\text{H}_2 + \text{CO}$. Meanwhile, the region labeled 1 corresponds to a weakly interacting intermediate that is formed by the partial dissociation of the CH bond in H_2CO . Although there are no potential barriers separating A from 1, there are entropic barriers that correlate with transition states separating the two. Similarly, there are centrifugal barriers separating 1 from P_1 . The region labeled 2 also corresponds to a weakly interacting intermediate, but now with the weakly interacting H closest to the other H. In this case, there are no potential barriers separating 2 from P_1 or P_2 but there are again centrifugal and entropic barriers. Notably, there is a potential barrier separating intermediate 1 from intermediate 2. The roaming transition state, which separates 1 from 2, should pass through this roaming saddle point.

The contour diagram in Figure 1 also illustrates minima corresponding to cis and trans HOCH. The following kinetic scheme is readily generalized to treat the kinetics of the roaming isomerization to these species as described later. However, for simplicity we focus here and, in our application below, on solely the roaming dynamics to produce $\text{H}_2 + \text{CO}$ and consider HOCH to be part of H_2CO .

It is important to note that we do not consider in this analysis the contribution to the formation of $\text{H}_2 + \text{CO}$ from the three-center tight transition state. We assume the latter contribution to be separate from that due to roaming, with the two contributions being additive. A detailed discussion of this separation between tight and roaming transition states for a range of reactions will be provided elsewhere.⁶ There it will be shown that this separability is generally appropriate and depends on the existence of second-order saddle points separating the first-order tight and roaming saddle points. For the cases of H_2CO and CH_3CHO the second-order saddle points lie more than 10 kcal/mol above the corresponding radical asymptotes. This implies that the assumed separability is appropriate to at least this energy. We note however that this assumed separation may not be valid in all cases because it does depend on the existence of a significant barrier separating the two saddle points. For the CH_3CCH and CH_3OOH cases we find no evidence for the existence of competing tight transition states.

Steady-State Kinetics. With this picture, one may write down the kinetic equations for the concentrations of species A, 1, 2, P_1 , and P_2 :

$$d[A]/dt = k_{1,A}[1] - k_{A,1}[A] \quad (1)$$

$$d[1]/dt = k_{A,1}[A] + k_{2,1}[2] - (k_{1,A} + k_{1,2} + k_{1,P_1})[1] \quad (2)$$

$$d[2]/dt = k_{1,2}[1] - (k_{2,1} + k_{2,P_1} + k_{2,P_2})[2] \quad (3)$$

$$d[\text{P}_1]/dt = k_{1,P_1}[1] + k_{2,P_1}[2] \quad (4)$$

$$d[\text{P}_2]/dt = k_{2,P_2}[2] \quad (5)$$

where $k_{i,j}$ is the rate coefficient for transformation from species i to species j .

Assuming steady state for the two intermediates, 1 and 2, then yields the following expression for the branching between roaming (P_2) and simple dissociation (P_1):

$$\frac{k_{P_2}}{k_{P_1}} = k_{2,P_2} / \left[k_{2,P_1} \left(1 + \frac{k_{1,P_1}}{k_{1,2}} \right) + k_{1,P_1} \left(\frac{k_{2,1} + k_{2,P_2}}{k_{1,2}} \right) \right] \quad (6)$$

where k_{P_1} and k_{P_2} are the effective rate constants for formation of products P_1 and P_2 , respectively, from A. The rate coefficients $k_{i,j}$ may be expressed in terms of the reactive fluxes N_{ij} for the transformation from species i to species j through the relation $k_{i,j} = N_{ij}/(h\rho_i)$, where ρ_i is the density of states for species i . Here, these reactive fluxes will be evaluated according to transition-state theory as the minimum in the number of states for appropriately defined dividing surfaces separating species i from species j .

The transformation from rate coefficients to reactive fluxes yields the following expression for the branching ratio

$$\frac{k_{P_2}}{k_{P_1}} = N_{2,P_2} / \left[N_{2,P_1} \left(1 + \frac{N_{1,P_1}}{N_{1,2}} \right) + N_{1,P_1} \left(1 + \frac{N_{2,P_2}}{N_{1,2}} \right) \right] \quad (7)$$

As an aside, we note that the densities of states do not enter into this expression, which is a generic result for the prediction of branching ratios. It arises from microscopic reversibility (i.e., $N_{ij} = N_{ji}$) and suggests that experimental measurements of branching ratios, as in the recent study of the photodissociation of acetaldehyde,³⁶ provide a particularly stringent test of transition-state theory.

Expressions for the individual rate coefficients k_{P_1} and k_{P_2} are also readily derived. However, these expressions are more complex, and our primary interest here is in the branching ratios, so these expressions are not provided here. Note though that an explicit prescription for obtaining the rate constants for a completely general case is given as follows.

Rapid Roaming Limit. It is useful to consider various limiting cases for this branching between roaming and bimolecular product formation. For example, in the limit where the flux through the roaming transition state, $N_{1,2}$, is large relative to the fluxes from 1 to P_1 and from 2 to P_2 , eq 7 simplifies to

$$\frac{k_{P_2}}{k_{P_1}} = \frac{N_{2,P_2}}{N_{2,P_1} + N_{1,P_1}} \quad (8)$$

In this rapid roaming limit, the two intermediates equilibrate and the branching is just given by the relative magnitude of the total fluxes to each of the products. This limit will be applicable when the roaming saddle point is well below the dissociation threshold and the energy is not too high.

Slow Roaming Limit. In the opposite slow roaming limit eq 7 reduces to

$$\frac{k_{P_2}}{k_{P_1}} = \frac{N_{1,2}N_{2,P_2}}{N_{1,P_1}(N_{2,P_1} + N_{2,P_2})} \quad (9)$$

Now, the rate to P_2 from intermediate 1 is essentially determined by the flux from intermediate 1 to 2, $N_{1,2}$, multiplied by the probability of product 2 formation from intermediate 2, $N_{2,P_2}/(N_{2,P_1} + N_{2,P_2})$, while the rate to P_1 from 1 is essentially determined by the flux from 1 to P_1 . This limit will be most applicable when the roaming saddle point is not very much below the dissociation threshold or the energy of interest is well above the threshold.

Rapid Abstraction Limit. Another interesting limit is obtained by considering the case where N_{2,P_2} is much larger than N_{2,P_1} . In this case, all roaming leads to abstraction and eq 7 reduces to

$$\frac{k_{P_2}}{k_{P_1}} = \frac{N_{1,2}}{N_{1,P_1}} \quad (10)$$

In this rapid abstraction limit, the overall branching is simply given by the ratio of the roaming flux to the product formation flux from the initially formed intermediate. This limit should be applicable at low energies.

Slow Abstraction Limit. In the opposite slow abstraction/rapid dissociation limit, with small N_{2,P_2} and large N_{1,P_1} , both relative to $N_{1,2}$, eq 7 reduces to

$$\frac{k_{P_2}}{k_{P_1}} = \frac{N_{2,P_2}N_{1,2}}{N_{1,P_1}N_{2,P_1}} \quad (11)$$

The applicability of this limit depends on the strength of the chemical interactions in the abstraction region of the potential.

2.2. Generalization to Arbitrary Numbers of Intermediates and Products. The results provided in eqs 7–11 are quite general and provide simple expressions for the most common roaming mechanism involving two intermediates and a competition between two products. However, more complicated situations that involve roaming between multiple long-range minima and cases where there are multiple products induced by roaming are also quite commonplace. For example, for formaldehyde dissociation, there are actually long-range minima that correlate with the roaming-induced production of HCOH.

The kinetic scheme is readily generalized to the case of an arbitrary number n of long-range minima and m products via analogy with the approach employed in our treatment of the collisionless limit rate constants for bimolecular reactions.³⁷ First, the time dependence of the concentration for each of the intermediates can be written as

$$d|I\rangle/dt = -\mathbf{K}|I\rangle + [\mathbf{A}][k_d] \quad (12)$$

where $|I\rangle = [|1\rangle, |2\rangle, \dots, |n\rangle]$ is a vector of concentrations for each of the intermediates, $|k_d\rangle = [k_{A,1}, k_{A,2}, \dots, k_{A,n}]$ is a vector of partial dissociation rate constants from the reactants A to each intermediate, and $\mathbf{K} = \mathbf{K}_{\text{isom}} + \mathbf{K}_{\text{prod}} + \mathbf{K}_{\text{rev}}$ is the sum of the rate coefficient matrices for isomerization, product formation, and reverse dissociation. The i th diagonal element of the isomerization matrix $\mathbf{K}_{\text{isom}}$ is given by $\sum_j k_{i,j}$, while the i,j th off-diagonal element is simply $-k_{i,j}$ with i and j both ranging from 1 to n . The product formation matrix $\mathbf{K}_{\text{prod}}$ is a diagonal matrix with elements given by $\sum_j k_{i,P_j}$, where now j ranges from 1 to m . The matrix $\mathbf{K}_{\text{rev}}$ is also diagonal with elements given by $k_{i,A}$. Applying the steady-state approximation to each of the intermediates yields

$$|I\rangle = [\mathbf{A}]\mathbf{K}^{-1}|k_d\rangle \quad (13)$$

where $\mathbf{K}^{-1}$ is the inverse matrix of $\mathbf{K}$. The time dependence of the product concentrations may be written as

$$d|P\rangle/dt = \mathbf{K}_p|I\rangle = [\mathbf{A}]\mathbf{K}_p\mathbf{K}^{-1}|k_d\rangle \quad (14)$$

where the i,j th element of the product formation matrix $\mathbf{K}_p$ is the rate coefficient k_{i,P_j} for forming product i from intermediate j . Transforming from rate constants to reactive fluxes then yields the following expression for the effective product formation rate coefficients, $|k_{\text{eff}}^P\rangle = |k_{\text{eff}}^{P_1}, k_{\text{eff}}^{P_2}, \dots, k_{\text{eff}}^{P_m}\rangle$,

$$|k_{\text{eff}}^P\rangle = \mathbf{N}_p\mathbf{N}^{-1}|k_d\rangle \quad (15)$$

where the $\mathbf{N}_p$ and $\mathbf{N}$ matrices are related to the $\mathbf{K}_p$ and $\mathbf{K}$ matrices, respectively, through substitution of the reactive fluxes for the rate coefficients. This expression is readily implemented numerically. For simple cases, such as the two-intermediate, two-product case considered in section 2.1, it is also readily solved analytically.

2.3. Generalization to Bimolecular Reactions. The generalization to bimolecular reactants, A and B, simply involves (i) the replacement of the $[\mathbf{A}][k_d]$ term in eq 12 with the corresponding bimolecular association rate constant $[\mathbf{A}][\mathbf{B}][k_a]$, where now $|k_a\rangle = [k_{A+B,1}, k_{A+B,2}, \dots, k_{A+B,n}]$ is the vector of association rate coefficients for forming each of the intermediates, and (ii) the replacement of $\mathbf{K}_{\text{rev}}$ for dissociation with $\mathbf{K}_{\text{rev}}$ for association whose diagonal elements are $k_{i,A+B}$. The intermediate concentrations are now given by

$$d|I\rangle/dt = -\mathbf{K}|I\rangle + [\mathbf{A}][\mathbf{B}][k_a] \quad (16)$$

The effective product formation rate coefficients are then given by

$$|k_{\text{eff}}^P\rangle = \mathbf{N}_p\mathbf{N}^{-1}|k_a\rangle \quad (17)$$

For the H + HCO reaction, this expression yields a branching ratio for the formation of H_2 + CO relative to formation of H_2CO given by

$$\frac{k_{P_2}}{k_{P_1}} = \frac{N_{2,P_2}[N_{1,2}N_{A+B} + (N_{1,P_1} + N_{A+B,1})N_{A+B,2}]}{N_{1,P_1}[N_{1,2}N_{A+B} + (N_{2,P_2} + N_{A+B,2})N_{A+B,1}]} \quad (18)$$

where N_{A+B} is the sum of the fluxes $N_{A+B,i}$ for formation of intermediate i from reactants A and B.

In the rapid roaming limit, large $N_{1,2}$, eq 18 reduces to

$$\frac{k_{P_2}}{k_{P_1}} = \frac{N_{2,P_2}}{N_{1,P_1}} \quad (19)$$

while in the slow roaming limit, small $N_{1,2}$, it reduces to

$$\frac{k_{P_2}}{k_{P_1}} = \frac{N_{2,P_2}(N_{1,P_1} + N_{A+B,1})N_{A+B,2}}{N_{1,P_1}(N_{2,P_2} + N_{A+B,2})N_{A+B,1}} \quad (20)$$

The quantity $N_{A+B,2}[N_{2,P_2}/(N_{2,P_2} + N_{A+B,2})]$ can be interpreted as the flux for forming intermediate 2, $N_{A+B,2}$, times the probability for proceeding on to product 2 from intermediate 2, $N_{2,P_2}/(N_{2,P_2} + N_{A+B,2})$, with a similar interpretation for the terms in intermediate 1 and product 1. Note that product 1 now denotes the molecular complex, e.g., H_2CO for $\text{HCO} + \text{H}$ as reactants.

The rapid product formation limit, with large N_{1,P_1} and N_{2,P_2} , yields

$$\frac{k_{P_2}}{k_{P_1}} = \frac{N_{A+B,2}}{N_{A+B,1}} \quad (21)$$

The opposite slow product formation limit, with large $N_{A+B,1}$ and $N_{A+B,2}$, yields eq 19 again.

3. DIVIDING SURFACES, STATE COUNTING, AND TRAJECTORY SIMULATIONS

3.1. Reduced Dimensional Framework and Variable Reaction Coordinate—Transition-State Theory. The implementation of the above-described statistical kinetics approach requires some procedure for estimating the transition-state fluxes, $N_{i,j}$, for each component of the dividing surfaces depicted in Figure 1. Notably, the transitional mode frequencies at the roaming saddlepoint range from about 20 to 100 cm^{-1} .^{4,5} For such low frequencies the intermolecular motions are of very large amplitude. Correspondingly, anharmonicities and mode couplings are expected to be significant and traditional rigid-rotor harmonic oscillator based approaches are not expected to be accurate.

Importantly, the intermolecular separations of the two reacting fragments in the regions that define the roaming intermediates are quite large. For example, the separations between the two radical sites are generally 2 Å or more for the inner transition states (e.g., those dividing surfaces separating A from intermediate 1 and P₂ from intermediate 2 in Figure 1). For such large separations the interactions between the two incipient radicals are relatively weak, and their vibrational modes are expected to behave adiabatically, with conserved quantum numbers. In this instance, a separation of modes into the conserved vibrational motions of the radicals and the remaining “transitional” modes is appropriate. The roaming dynamics involves predominantly the transitional modes.

In our efforts to predict the rate coefficient for the recombination of two radicals, we have developed a transition state theory (TST) approach that provides an accurate treatment of the anharmonicities and mode couplings for the transitional modes.^{27,38,39} The separation into conserved and transitional modes provides the foundation for this approach, with the conserved modes treated via direct sums over the quantum eigenstates (typically evaluated at the harmonic oscillator level), while the transitional modes are evaluated via classical phase space integrals. The latter classical evaluations are appropriate due to the low-frequency nature of the transitional modes.

This approach also includes a variable definition of the reaction coordinate, whose optimization is an important aspect of accurately predicting the reactive flux. In this variable reaction coordinate (VRC) approach, the transitional modes are defined in terms of rotations of each of the fragments about fixed pivot points and of the intermolecular axis connecting the pivot points. The variational component of the calculations then involves minimization of the calculated number of states with respect to the fixed separation between the pivot points and also with respect to the pivot point locations. Fairly general dividing surfaces can be obtained via the consideration of an appropriate set of pivot points on each of the fragments.

3.2. Planar Dividing Surfaces. Here we employ a generalization of the VRC–TST approach as a means to evaluate the reactive fluxes, N_{ij} . The consideration of fixed pivot-to-pivot separations yields spherical dividing surfaces. Such spherical dividing surfaces are well-suited for consideration of the inner and outer transition-state dividing surfaces represented by the nearly circular lines shown in Figure 1. However, such spherical dividing surfaces do not appear to be particularly appropriate for the roaming transition states, which are depicted as straight lines in Figure 1. Instead, it would appear that planar dividing surfaces provide a more reasonable approximation for roaming transition states.

In a previous paper,⁴⁰ we have derived the expression for the reactive flux through a planar dividing surface when the other fragment is an atom. In this section we generalize this expression to partner fragments that are either linear or nonlinear. The dividing surface for the nonlinear case can be defined by the following equation,

$$s = \mathbf{n}^{(0)} \cdot (\mathbf{r} + \mathbf{d}) \quad (22)$$

where $\mathbf{n}^{(0)}$ is the normal to the plane associated with the first fragment, $\mathbf{r}$ is the vector connecting the centers of mass of the fragments, and $\mathbf{d}$ is the vector connecting the center of mass of the second fragment and the pivot point associated with it. The

value of the reaction coordinate s defines the shift of the plane relative to the center of mass of the first fragment.

Taking the time derivative of eq 22 one obtains the following expression for $\dot{s}$ in terms of the renormalized orbital momentum, $\mathbf{p} \rightarrow \mathbf{p}/\sqrt{\mu}$, and the renormalized angular momenta of the fragments, $\mathbf{J}_i^{(k)} \rightarrow \mathbf{J}_i^{(k)}/\sqrt{I_i^{(k)}}$,

$$\begin{aligned} \dot{s} = & \mu^{-1/2} \mathbf{n}^{(0)} \cdot \mathbf{p} - \sum_{i=1}^3 \frac{(\mathbf{n}_i^{(2)}, \mathbf{n}^{(0)}, \mathbf{d})}{\sqrt{I_i^{(2)}}} J_i^{(2)} \\ & + \sum_{i=1}^3 \frac{(\mathbf{n}_i^{(1)}, \mathbf{n}^{(0)}, \mathbf{r} + \mathbf{d})}{\sqrt{I_i^{(1)}}} J_i^{(1)} \end{aligned} \quad (23)$$

where μ is the reduced mass of the reaction complex, $\mathbf{n}_i^{(k)}$ and $I_i^{(k)}$ are the unit vector and the moment of inertia associated with the i th principal axis of the k th fragment, and $(a, b, c) \equiv (a \times b) \cdot c$. The coefficients in front of $\mathbf{p}$ and $J_i^{(k)}$ in eq 23 provide the components of the ν -dimensional reaction coordinate vector $\mathbf{X}$ in the generalized momenta vector space; cf. eq 2.15 of ref 40. The knowledge of this vector is important for constructing initial conditions in the classical trajectory calculation of the recrossing factor and for the statistical calculation of the reactive flux through the dividing surface. In particular, the kinematic factor Φ , which comes into the expression for the number of states $N(E, J)$ (cf. eq 2.39 of ref 40) is given by

$$\begin{aligned} \Phi & \equiv \sqrt{\mu \mathbf{X} \cdot \mathbf{X}} \\ & = \sqrt{1 + \mu \sum_{i=1}^3 \frac{(\mathbf{n}_i^{(2)}, \mathbf{n}^{(0)}, \mathbf{d})^2}{I_i^{(2)}} + \mu \sum_{i=1}^3 \frac{(\mathbf{n}_i^{(1)}, \mathbf{n}^{(0)}, \mathbf{r} + \mathbf{d})^2}{I_i^{(1)}}} \end{aligned} \quad (24)$$

In the case when the second fragment is a linear molecule, eqs 23 and 24 are modified as follows,

$$\begin{aligned} \dot{s} = & \mu^{-1/2} \mathbf{n}^{(0)} \cdot \mathbf{p} - \frac{(J^{(2)}, \mathbf{n}^{(0)}, \mathbf{d})}{\sqrt{I^{(2)}}} \\ & + \sum_{i=1}^3 \frac{(\mathbf{n}_i^{(1)}, \mathbf{n}^{(0)}, \mathbf{r} + \mathbf{d})}{\sqrt{I_i^{(1)}}} J_i^{(1)} \end{aligned} \quad (25)$$

$$\Phi = \sqrt{1 + \frac{\mu}{I^{(2)}} d^2 \sin^2 \theta + \mu \sum_{i=1}^3 \frac{(\mathbf{n}_i^{(1)}, \mathbf{n}^{(0)}, \mathbf{r} + \mathbf{d})^2}{I_i^{(1)}}} \quad (26)$$

where θ is the angle between the normal to the plane associated with the first fragment and the symmetry axis of the second fragment.

The kinematic factor characterizes the configurational contributions to the reactive flux from different types of motion without taking into account the dynamical factor (potential energy term). It is instructive to estimate the kinematic factor for a plane that separates different angular sectors. For estimation purposes one can assume that the dividing plane passes through the center of mass of the first fragment and that the pivot point coincides with the center of mass of the second fragment, $d \approx 0$. Then $\mathbf{n}^{(0)} \perp \mathbf{r}$, and at large interfragment distances the main contribution in eq 24 comes from the last term, which is

associated with rotation of the first fragment, and the kinematic factor can be estimated as, $\Phi \sim (\mu r^2/I^{(1)})^{1/2} \gg 1$.

Now, it is instructive to contrast this result with the typical values for our variable reaction coordinate (spherical) dividing surfaces. Notably, for the center-of-mass to center-of-mass reaction coordinate $\Phi = 1$. As the fragment pivot points move away from the center-of-mass, the kinematic factor increases. The optimal reaction coordinate then arises from some balance of this increase in the kinematic factor with the decrease in the average potential energy term (which can be viewed qualitatively as the configurational integral of the Boltzmann factor). At large separations the potential energy term is close to unity and the optimal reaction coordinate is the center-of-mass to center-of-mass separation. At shorter separations the potential energy term deviates significantly from unity and the pivot points move away from the center-of-mass in an attempt to minimize the potential energy term.

For the roaming transition states, the optimal potential energy terms are obtained for planar dividing surfaces. Furthermore, even modest deviations from a plane significantly increase the contributions from large separations due to an increase in the configurational volume. Thus, it seems reasonable to presume that planes provide the optimal dividing surface, even though their kinematic factors deviate significantly from unity.

3.3. General Dividing Surfaces. The proper separation of the reactants, intermediates, and products, will generally require the consideration of contributions from both planar and spherical dividing surfaces. To see this, consider Figure 1 and imagine a case where the optimal spherical dividing surface separating A from 1 is of smaller radius than that separating P2 from 2. In this case, the flux into P₂ would consist of a spherical dividing surface contribution from 2 to P₂ and a planar dividing surface contribution from 1 to P₂. As the roaming dynamics gets more complex, the number of contributions increases, and the bookkeeping gets rather involved.

As part of this work, we have developed a general method that allows one to evaluate the reactive fluxes between different parts of configurational space representing different chemical species, both unimolecular and bimolecular. The key feature of the method is that it allows one to consider an arbitrary number of configurational space regions of quite general shape and to efficiently sample the boundaries between them. A set of predefined elementary surfaces, which currently include spherical shapes and planes but can be easily extended to different shapes, is used to define chemical species. Each elementary surface divides the whole configurational space into two parts, one of which can be considered as an elementary subspace associated with this surface and the other one as the supplement/negation of the first one. For example, for the spherical surface the elementary subspace is defined as an inner volume of the sphere and for the planar surface the corresponding subspace includes all configurations in the direction of the normal to that plane.

A chemical species is then defined in terms of binary logic as an arbitrary combination of unions, intersections, and negations between different elementary subspaces. In this manner the whole configurational space is subdivided into several nonintersecting regions representing different chemical species. A random sampling of all elementary surfaces is performed and the efficient algorithm allows one to determine if the current sampling belongs to the boundary between different species (and which species) or to the inner region of some species. The

contributions from random samplings associated with the specific elementary surface and the specific pair of chemical species are then summed up, and the fluxes between each of the species are evaluated including each of the contributions from all of the different pieces of all elementary components of the dividing surface. These fluxes are precisely what is required for the evaluation of the branching ratios according to the kinetic scheme outlined in section 2. The fluxes are evaluated to a given accuracy via Monte Carlo sampling of initial configurations. The preceding described method has been implemented into a computer code (CROSSRATE) that also includes classical trajectory propagations; see the following text.

3.4. Reduced Dimensional Trajectories. A reduced dimensional trajectory approach for studying the branching to roaming was previously described in our study of roaming in acetaldehyde decomposition.⁴ The basis of this approach is again the separation into the conserved and transitional modes, with the two fragments treated as rigid bodies during the trajectory propagation. Related rigid body trajectory simulations have been employed in a number of studies of ion–molecule^{41–44} and radical–radical^{45–49} addition reactions.

The trajectories are propagated forward and backward from an initial dividing surface in order to determine the species to species reactive fluxes for statistical distributions of initial states. The initial dividing surface, which is used for initiating the trajectories and for distinguishing between reactants and products upon termination of the trajectories, is again represented in terms of an arbitrary combination of spheres and planes. The trajectories are terminated when either large negative potential values are encountered or when the trajectory enters the region that is deemed as a terminal one and from which no return is assumed. Such terminal regions are defined in a similar manner as the initial dividing surface. In particular, the bimolecular terminal region is defined in terms of the maximal distance between the centers of masses of two radicals.

The CROSSRATE code again handles all the bookkeeping to evaluate all of the species to species reactive fluxes for all of the species defined by the given set of spherical and planar dividing surfaces. Random sampling of initial configurations is again used to obtain convergence to preset accuracy limits. Trajectory propagation is performed via an Adams–Bashforth–Moulton predictor–corrector algorithm with user defined accuracy limits. The code produces both forward and reverse reactive fluxes, which by microscopic reversibility should be identical. This equivalence provides a valuable test for the simulations.

4. POTENTIAL ENERGY SURFACES

Both the TST calculations and the trajectory simulations require a global potential energy surface describing the interaction of the two incipient radicals for arbitrary orientations at separations ranging from about 2 to about 10 Å. We have previously developed global analytic potentials for these transitional modes from direct fits to extensive grids of multireference electronic structure determined energies for a variety of reactions including H and CH₃ with HCO^{1,4,50} and H with C₃H₃.⁵¹ For this work, we develop a new potential for the CH₃O + OH reaction and make minor revisions (primarily in the active space and/or basis set employed) to these earlier potentials for the other three reactions.

For H + HCO the present potential energy surface is based on (12e,10o) CASPT2/aug-cc-pVDZ calculations. The (12e,10o)

active space correlates with the full-valence space. For $\text{CH}_3 + \text{HCO}$, we simply employ the potential energy surface from ref 4, which was based on ab initio calculations performed at the (2e,2o) CASPT2/aug-cc-pVDZ level. This potential also includes one-dimensional corrections for geometry relaxation, zero-point energy changes, and limitations in the active space and the basis set. For $\text{H} + \text{C}_3\text{H}_3$, the energies are obtained from (6e,6o) CASPT2/aug-cc-pVDZ calculations. The active space consists of the π -space of C_3H_3 together with the H radical orbital.

For $\text{CH}_3\text{O} + \text{OH}$ the present potential surface is based on (8e,6o)-CASPT2/aug-cc-pVDZ calculations. The geometries of both radicals are fixed at equilibrium. For CH_3O this corresponds to the Jahn–Teller distorted, $^2A'$, ground state. The active space includes the two singly occupied, oxygen-centered radical orbitals, the two doubly occupied, oxygen p lone pair orbitals and one CH (σ , σ^*) pair (the one in the CH_3O symmetry plane that is aligned with the CH_3O radical orbital). The CASSCF calculation was state-averaged over the four lowest electronic states, corresponding to the two degenerate components of the OH, $^2\Pi$, and the two nearly degenerate, Jahn–Teller components of the CH_3O . Electronic structure calculations were carried out at $\sim 60\,000$ geometries and the resulting energies fit with a segmented, Morse-variable type fit as described in ref 4.

Our recent roaming dynamics studies^{4,18} have suggested that various correction terms to the potential have a significant effect on the quantitative roaming branching fraction predicted. The conserved mode relaxation effects in the abstraction channels were particularly important. Corrections due to basis set and active space limitations and zero-point energy changes were also considered. Here our focus is on the comparison of our statistical kinetics picture with direct trajectory simulations. These correction terms should be of little consequence to this comparison, so, for simplicity, we have chosen to neglect them in the present analysis, aside from the $\text{CH}_3 + \text{HCO}$ case, where they were already evaluated and incorporated into the prior potential. Qualitatively, we expect their inclusion would only yield a modest increase in the amount of roaming at higher energies.

Note that no consideration is given here to the tight transition states that lead to a variety of products including the roaming products. Our focus is only on understanding the competition between dissociation and roaming in the long-range region. The contribution to the kinetics and dynamics from any tight transition states is generally well-separated from that due to roaming. In particular, the two contributions are simply additive. Detailed discussion and illustration of this separability will be provided in a forthcoming study.⁶

5. RESULTS AND DISCUSSION

5.1. H_2CO . As a first example of the application of the present statistical theory based kinetic scheme, we consider the dissociation of H_2CO into either $\text{HCO} + \text{H}$ (P_1) or $\text{H}_2 + \text{CO}$ (P_2). It is instructive to first consider the variation in the VRC–TST predicted reactive fluxes with reaction coordinate for the inner and outer transition states from the intermediate 2 ($\text{H} \cdots \text{HCO}$) in the dissociation of H_2CO . These fluxes are illustrated in Figure 2 for a range of energies (0.1, 0.3, 1.0, and 3.0 kcal/mol) with $J = 1$. The inner transition state separates $\text{H}_2 + \text{CO}$ from other species, primarily $\text{H} \cdots \text{HCO}$ (intermediate 2) but also $\text{H} \cdots \text{CHO}$ (intermediate 1) via the planar segments. Meanwhile the outer transition state separates $\text{HCO} + \text{H}$ from $\text{H} \cdots \text{HCO}$.

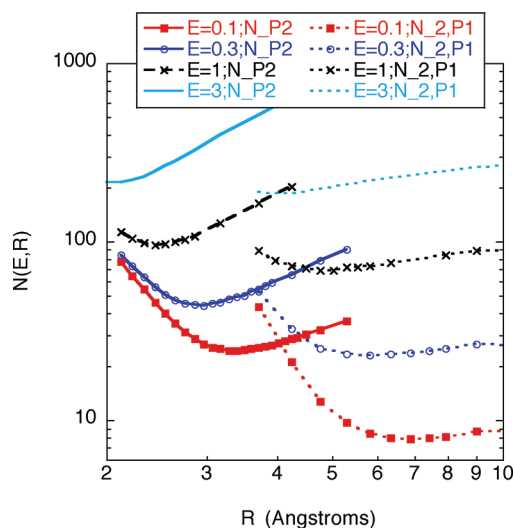


Figure 2. Plot of the reaction coordinate dependence of the energy E and total angular momentum J resolved reactive fluxes for the formation of P_2 (N_{P_2}) and for the formation of P_1 from intermediate 2 (N_{2,P_1}). Both fluxes are plotted for energies of 0.1, 0.3, 1.0, and 3.0 kcal/mol relative to $\text{HCO} + \text{H}$, and are for $J = 1$. For N_{P_2} , the coordinate R denotes the distance of the radical H atom from the H atom of HCO . For N_{2,P_1} , it denotes the center-of-mass to center-of-mass separation distance. The minima in these fluxes correlate with the optimal inner (N_{P_2}) and outer (N_{2,P_1}) transition states for intermediate 2.

These plots clearly illustrate the two transition state picture that forms the basis of the present kinetic scheme for predicting the branching to roaming.

The various minima in these plots correspond to the location of the transition state for the given energy and process. At low energies there is a clear distinction between the inner and outer transition states, with the flux at the inner transition state greatly exceeding that for the outer transition state due to the presence of significant attractive energies on the inner transition state dividing surface. The roaming saddle point energy is 0.2 kcal/mol below the $\text{H} + \text{HCO}$ asymptote. With increasing energy the distinction between the two decreases both in terms of the location and the reactive flux. Note that in comparing locations one should shift that for the inner transition state up by about 2 Å since the HCO pivot point is displaced from the center of mass of HCO by that amount.

At even higher energies, which are not plotted here, the distinction between the two completely disappears and the present scheme is not applicable. However, in this limit the roaming contribution is expected to be very small. Furthermore, the distinction between the inner and outer transition states persists to higher energies for systems that involve methyl or other larger roaming fragments rather than an H atom.

The energy dependence of the VRC–TST predictions for the four reactive flux components required for the evaluation of eq 7 are plotted in Figure 3. For this reaction, these four reactive fluxes are of comparable magnitude throughout the energy range plotted (0.05–7 kcal/mol). At low energies, the two components $N_{1,2}$ and N_{2,P_2} become increasingly dominant, while at high energies the reverse is true with the N_{1,P_1} and N_{2,P_1} components becoming dominant. This result, which arises directly from the negative energies on the inner transition-state dividing surface, suggests that the large roaming and large abstraction limits may be applicable at low energy.

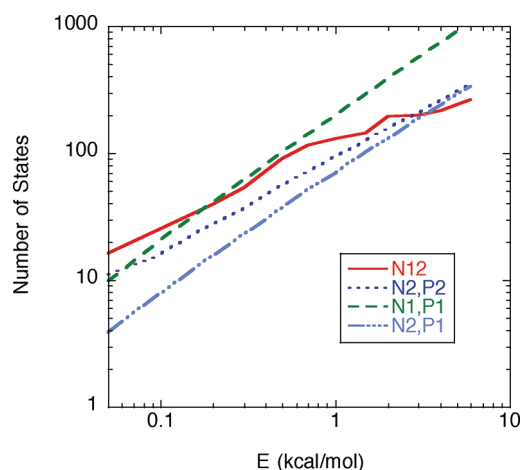


Figure 3. Plot of the TST predictions for the flux from intermediate 1 to intermediate 2 ($N_{1,2}$), from intermediate 1 to product 1 ($N_{1,P1}$), and from intermediate 2 to products 1 ($N_{2,P1}$) and 2 ($N_{2,P2}$). All plots are for the dissociation of H_2CO at the specified energy E relative to $\text{H} + \text{HCO}$ and for a total angular momentum $J = 1$.

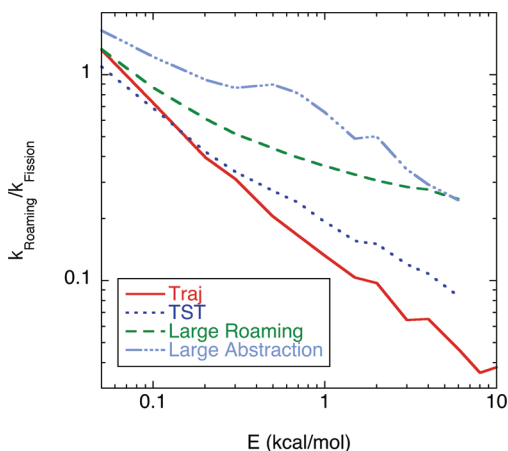


Figure 4. Plot of the energy dependence of the predicted branching to roaming in H_2CO decomposition as predicted with a variety of methods including trajectory simulations (Traj), the overall statistical kinetics picture (eq 7, TST), and the large roaming (eq 8) and large abstraction limits (eq 10). Results are for a total angular momentum $J = 1$.

The predictions for the roaming branching from the present kinetic scheme (eq 7) are contrasted with the results from trajectory simulations in Figure 4. The two results agree to within a factor of 2 over the full energy range and are within 20% for energies up to 0.4 kcal/mol. The rapid roaming limit (eq 8) also agrees well at low energies but then becomes increasingly discordant with increasing energy and is more than a factor of 5 too high at the highest energies. The large abstraction limit (eq 10) predicts too much roaming, with this overprediction persisting to low energies.

5.2. CH_3CHO . The decomposition of acetaldehyde is analogous to that of formaldehyde but with the methyl radical roaming instead of the H atom. A contour plot of the $\text{CH}_3 + \text{HCO}$ interaction potential is provided in Figure 5. The strongly attractive contours on the top and bottom right correlate with $\text{CH}_4 + \text{CO}$ and CH_3CHO , respectively. The saddle point on the middle right of the plot correlates with the roaming saddle point for motion in the plane plotted. However, this saddle point is actually a

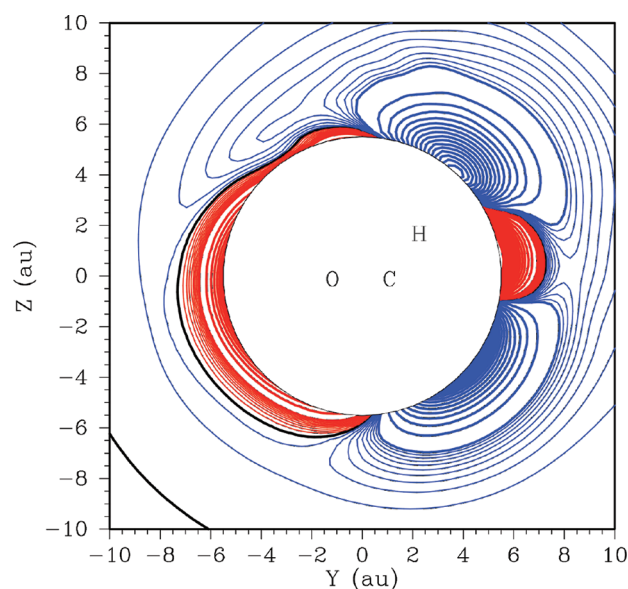


Figure 5. Contour plot of the interaction of the methyl radical with a fixed HCO radical. The blue and red contours denote attractive and repulsive interactions, respectively. The thin lines have a contour spacing of 0.1 kcal/mol, while the thick lines have a spacing of 1.0 kcal/mol. The distances are in atomic units.

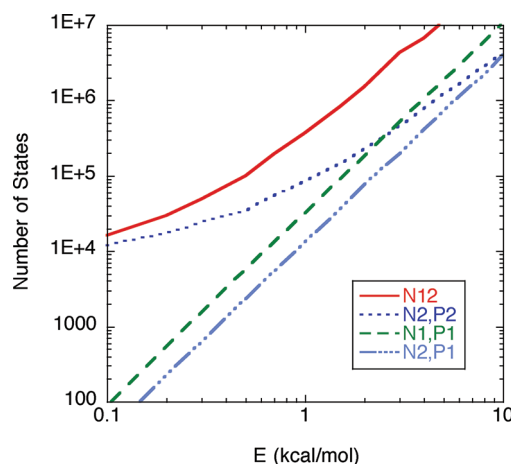


Figure 6. As in Figure 3, but all plots are for the dissociation of CH_3CHO at the specified energy E relative to $\text{CH}_3 + \text{HCO}$ and for a total angular momentum $J = 1$ (where, for example, $1\text{E}+7$ represents 1×10^7).

second-order saddle point and the minimum energy roaming pathway involves an out of plane roaming motion with a roaming saddle point at an energy of 1.3 kcal/mol below the $\text{CH}_3 + \text{HCO}$ asymptote. Actually, the roaming pathway and saddle point for $\text{H} + \text{HCO}$ are also out of the plane of the HCO radical.

The VRC-TST predictions for the energy dependence of the key components of the reactive flux are illustrated in Figure 6 for $J = 1$. This time, the dominance of the roaming and abstraction fluxes ($N_{1,2}$ and $N_{2,P2}$, respectively) at low energy is clear. Again, this low-energy dominance is due to the negative energies for their dividing surfaces. At higher energies the fluxes are all of similar magnitude, although the roaming flux continues to be the largest.

As illustrated in Figure 7, the statistical kinetic predictions for the branching to roaming are again in quantitative agreement

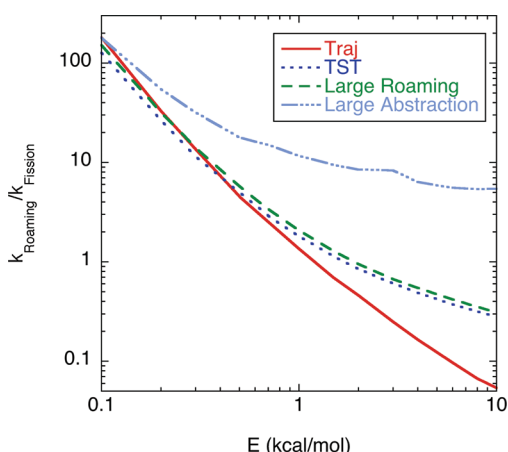


Figure 7. As in Figure 4, but plots are for the dissociation of CH_3CHO with $J = 1$.

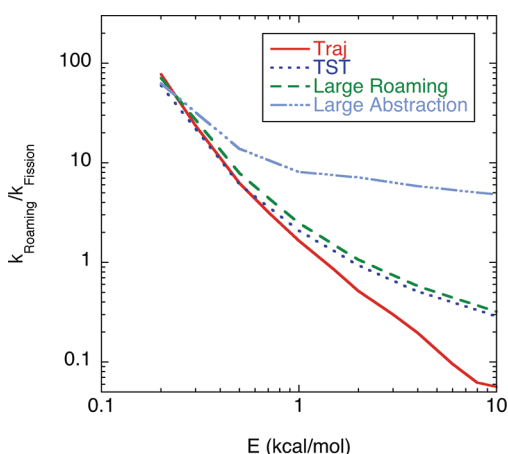


Figure 8. As in Figure 7, but for a total angular momentum $J = 25$.

with the trajectory simulations at low energies. The agreement remains good until about 1 kcal/mol, at which point the statistical kinetic predictions increasingly overestimate the trajectory results. At 10 kcal/mol, which is the highest energy considered, the statistical kinetic scheme overpredicts the roaming by about a factor of 6.

Interestingly, the predictions based on the rapid roaming limit agree quantitatively with the full kinetic scheme results over the full range of energy from 0.1 to 10 kcal/mol. The increased applicability of this limit, in comparison with formaldehyde dissociation, is related to the lowering of the roaming saddle point energy from -0.2 to -1.3 kcal/mol. In contrast, the large abstraction limit is still of only limited applicability, i.e., to the very low energy limit.

The corresponding plots in Figure 8 of the statistical kinetic and trajectory results for $J = 25$ demonstrate that the roaming process is not strongly dependent on angular momentum.

The key assumption in the present statistical model is one of a randomization of the transitional mode energy within the long-range minimum. The discrepancy at high energies between the statistical kinetic predictions and the trajectory results is likely indicative of a dynamical biasing toward continued dissociation from intermediate 1 on to $\text{CH}_3 + \text{HCO}$ for trajectories that enter the intermediate 1 region with a high radial kinetic energy.

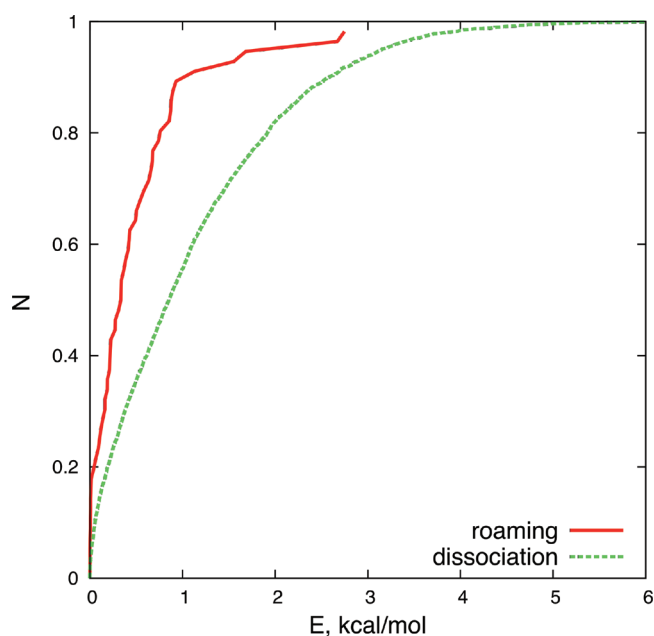


Figure 9. Plot of the fraction of trajectories that have a given radial kinetic energy for the trajectories leading to either roaming or dissociation. Results are for an energy of 4 kcal/mol relative to $\text{CH}_3 + \text{HCO}$.

Trajectories that have a high radial kinetic energy may not be effectively slowed by the centrifugal barriers of the outer transition state. The probability of such high radial kinetic energy trajectories increases with increasing total energy, so the statistical kinetic picture gradually breaks down. The convergence of the roaming and radical flux with radial kinetic energy at the transition state is plotted in Figure 9. The fact that the roaming flux converges at much lower radial kinetic energy correlates with the above-mentioned dynamical biasing.

In principle, one could account for this dynamical biasing by presuming that any flux crossing the dividing surface from A to 1 with a radial kinetic energy exceeding some threshold (e.g., 1 kcal/mol or even better some orbital angular momentum dependent threshold) proceeds directly to products. In essence, one would modify the kinetic scheme to include a contribution from an A to P_1 flux. We have not done so here.

5.3. CH_3OOH . A contour plot of the interaction potential for $\text{OH} + \text{CH}_3\text{O}$ is provided in Figure 10. The strong attractions at the top and bottom left of the plot correlate with different torsional states of CH_3OOH . The strong attraction at the top right corresponds to abstraction of one of the H atoms in CH_3O by the OH group. The roaming saddle point is 1.9 kcal/mol below the $\text{CH}_3\text{O} + \text{OH}$ asymptote and lies at the top middle of the plot. A vertical plane passing through this saddle point provides a reasonable approximation to the roaming transition-state dividing surface.

The VRC-TST predictions for the component fluxes are illustrated in Figure 11. Again, we see a dominance of the roaming and abstraction fluxes at low energy and a similar magnitude for all the fluxes at high energy.

The statistical kinetic predictions for the branching to roaming are compared to the trajectory results in Figure 12. The picture is similar to that for acetaldehyde, with good agreement between the full statistical predictions and trajectory results at low energy and the applicability of the rapid roaming limit throughout.

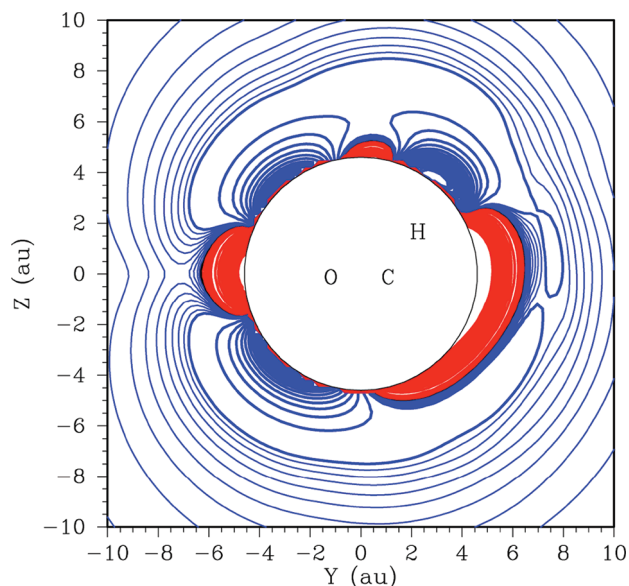


Figure 10. Contour plot of the interaction of the OH radical with a fixed methoxy radical. The out of plane H atoms on CH_3O are not illustrated. The blue lines are attractive, while the red are repulsive. The thin lines have a spacing of 0.1 kcal/mol, while the thick lines have a spacing of 1.0 kcal/mol.

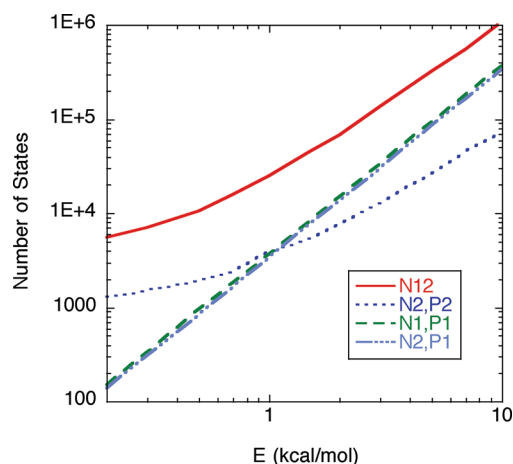


Figure 11. As in Figure 3, but the results are for the dissociation of CH_3OOH with $J = 1$ (where, for example, $1\text{E}+6$ represents 1×10^6).

In this case, though, the discrepancy between the statistical kinetic and trajectory results starts at somewhat lower energies and is even greater at higher energies. Note, however, that the discrepancies are more similar for the two molecules if one focuses on the branching ratio rather than E . In particular, the two curves start to diverge for both reactions when the roaming and fission rate constants are about equal; i.e., the branching = 1. It is just that the roaming branching at a given E is lower for CH_3OOH .

5.4. C_3H_4 . As a final example we consider the branching between dissociation and isomerization in propyne. A contour plot illustrating the interaction of an H atom with a fixed propargyl radical is provided in Figure 13. The strongly attractive regions on the left correlate with allene, while those on the right correlate with propyne. The roaming saddle point, which connects propyne to allene, appears in the middle of the plot at both

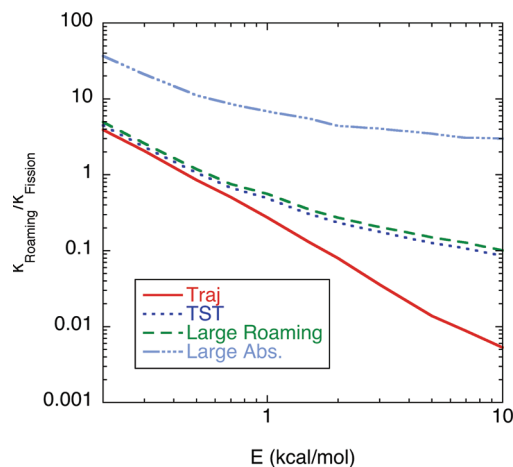


Figure 12. As in Figure 4, but the results are for the dissociation of CH_3OOH with $J = 1$.

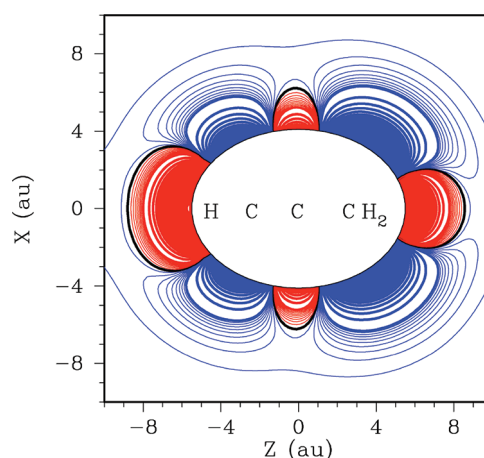


Figure 13. Contour plot of the interaction of an H atom with a fixed propargyl radical. The H atoms on the CH_2 side of the propargyl radical are coming out of the plane. The blue lines are attractive, while the red are repulsive. The thin lines have a spacing of 0.1 kcal/mol, while the thick lines have a spacing of 1.0 kcal/mol.

the top and the bottom. This saddle point lies 0.2 kcal/mol below the $\text{H} + \text{propargyl}$ asymptote. Notably, the roaming saddle points for the two H roaming cases are quite similar, but are much less attractive than those for the CH_3 and OH roaming cases.

The VRC–TST predictions for the reactive flux components are illustrated in Figure 14. Again, the fluxes for roaming and for intermediate 2 to product 2, which is isomerization in this case, are dominant at low energy. However, in this case the isomerization flux actually dominates over the roaming flux.

The statistical kinetic predictions for the roaming branching are illustrated in Figure 15 along with the trajectory results. Note that the scale of the x -axis for this figure is 0.01–1 kcal/mol, i.e., 10 times lower than for the other figures. The results are fairly similar to those for formaldehyde decomposition but with the large isomerization limit applicable over a somewhat broader energy range.

5.5. Generalizations. The expression for the branching to roaming given in eq 7 indicates that the branching to roaming products will be greatest when (i) the flux N_{2,P_2} from the roaming-induced intermediate to the roaming products is large

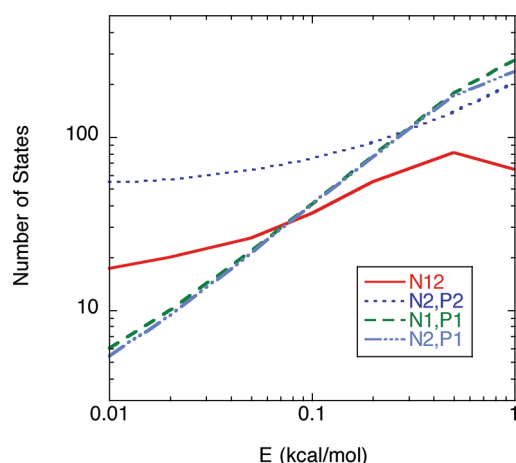


Figure 14. As in Figure 3, but for the dissociation of CH_3CCH with $J = 1$.

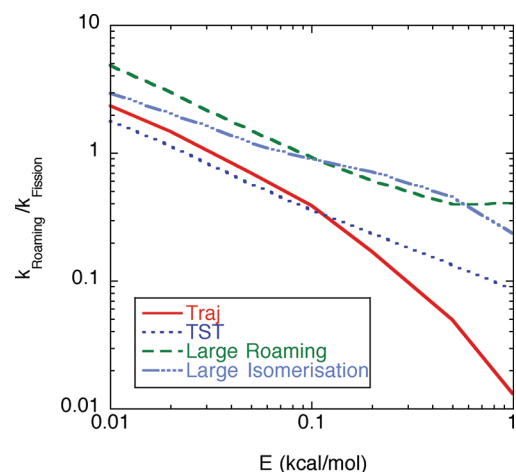


Figure 15. As in Figure 4, but for the dissociation of propyne with $J = 1$.

and (ii) the roaming flux $N_{1,2}$ connecting the long-range intermediates is large relative to other fluxes. The branching between abstraction and addition for radical–radical reactions provides some indication of the first factor. When this branching is small, one expects little roaming in the corresponding dissociation. Notably, for $\text{CH}_3 + \text{HCO}$ and $\text{H} + \text{HCO}$, where roaming has clearly been observed, this branching approaches unity.⁵

The relative magnitude of the roaming flux is strongly dependent on the energy of the corresponding saddle point relative to separated fragments. Indeed, for the range of energies from the roaming saddle point to the dissociation threshold only roaming reactions can occur. Notably, the strength of the long-range interactions seems to have a strong affect on the energy of this saddle point. For ion–molecule reactions, the long-range interactions are particularly strong and roaming is commonly a dominant aspect of the kinetics. For radical–radical reactions the dispersion coefficient gradually increases with molecular size and correspondingly one might expect a gradual lowering of the roaming saddle point energy relative to radical fragments. For example, for the series of aldehydes, CH_3CHO , $\text{CH}_3\text{CH}_2\text{CHO}$, $(\text{CH}_3)_2\text{CHCHO}$, and $(\text{CH}_3)_3\text{CCHO}$ we calculate roaming saddle points for the alkyl group roaming about HCO at -1.2 , -2.3 , -3.2 , and -3.9 kcal/mol, relative to separated radicals.

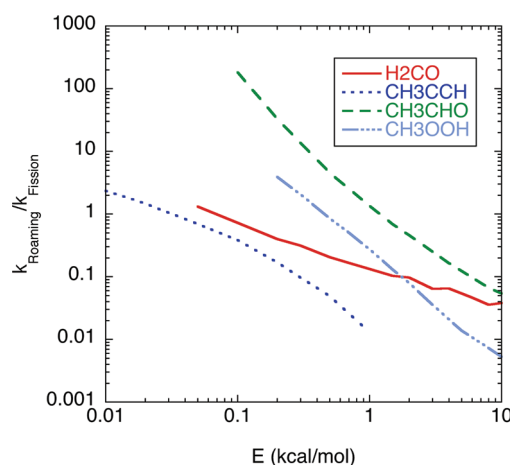


Figure 16. Plot of the energy dependence of the predicted branching to roaming for the dissociations of H_2CO , CH_3CCH , CH_3CHO , and CH_3OOH . All plots are for a total angular momentum $J = 1$.

However, for the alkanes, the roaming saddle points for propane, isobutane, and neopentane are at -1.2 , -1.5 , and -0.9 kcal/mol, respectively. The value for neopentane deviates from the expected trend, perhaps due to some steric effect. For H atoms the dispersion coefficients are much smaller, and the long-range interactions are much weaker, with roaming saddle points typically only a few tenths of a kilocalorie per mole below fragments. Interestingly, the hydrogen bonds that are prevalent in species containing N and O atoms are also likely to lead to lower roaming saddle point energies.

In Figure 16 we plot the results of the present trajectory simulations of the roaming branching for all four molecules studied here. The H_2CO and CH_3CCH dissociations, which involve H atom roaming, are quite similar, while the CH_3CHO and CH_3OOH dissociations, which involve CH_3 and OH roaming, respectively, are quite similar. The increased roaming for the latter two reactions correlates well with the increased attractiveness of their roaming saddle points.

6. CONCLUDING REMARKS

We have presented a statistical picture for the kinetics and dynamics of roaming reactions. This formalism is based on the statistical treatment of the kinetics of long-range effective intermediates. The presence of inner and outer transition states for the dissociation is central to the idea that such intermediates exist, at least dynamically. The roaming transition states, which pass through roaming saddle points, describe the isomerization kinetics for these long-range intermediates. A separation into the “conserved” internal modes of the incipient fragments and the “transitional” intermolecular modes allows for the quantitative implementation of this statistical picture through generalizations of our variable reaction coordinate–transition state theory approach.

Direct comparisons with trajectory simulations indicate that, at least within the transitional mode subspace, this statistical picture is accurate at low energies and then gradually overestimates the roaming branching as the energy increased. This overprediction appears to be related to a dynamical bias toward continued dissociation for trajectories that cross the inner transition state with high radial kinetic energy. Typically, the statistical theory and trajectory results begin to diverge at an

energy of about 1 kcal/mol, or more precisely when the roaming and decomposition branching are more or less equivalent. From an applied perspective, a modest overestimate of the roaming branching at high energies may not be too problematic, as the contribution from roaming is already quite small there.

The formalism provides a useful indication of the key factors that affect the roaming branching. Roaming saddle points that are low in energy relative to the separated fragments will yield more extensive roaming. Correspondingly, due to dispersion and other long-range potential effects, the roaming tends to be significantly greater for molecular radicals than for H atoms. Furthermore, when the related bimolecular abstraction rates are competitive with the corresponding bimolecular addition rates, one can expect the branching to roaming to be relatively large.

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